

Control Network Construction for LRO NAC Images Based on Refining Tie Points by Matching With Shaded LOLA DEM

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Abstract—The upcoming exploration activities in the lunar south pole require high-resolution mapping products to support landing site selection and navigation. The special terrain and illumination conditions in the lunar south pole pose significant challenges to photogrammetric processing. Existing lunar mapping products exhibit limited coverage and accuracy in the south pole region and cannot meet the exploration missions' requirements. In this study, a new control network construction method is proposed based on refining tie points by image matching between the approximate orthophotos of lunar reconnaissance orbiter narrow-angle camera (NAC) images and shaded digital elevation model (DEM) maps. The shaded DEM maps are generated using lunar orbiter laser altimeter terrain models with the same acquisition time of on-orbit NAC images, making them have very similar illumination conditions to the on-orbit images. Using the shaded DEM maps as a transition, we establish the coordinate transformation relationship between the stereopairs of NAC images by computing the homograph transformation matrix in orthophoto space. Then, the candidate control measures in the control network are refined in orthophoto space and then converted to the original image space. We conducted experiments using two sets of NAC images near the lunar south pole. The experimental results demonstrate that the proposed method can effectively solve the problems of image matching and control network generation for lunar south pole images.

Index Terms—Bundle adjustment, control network, image matching, planetary photogrammetry.

I. INTRODUCTION

THE lunar south pole (LSP) is the focus of current deep space exploration activities. Major space agencies have implemented or plan to conduct various exploration activities in the LSP, including the Artemis-3 lunar landing and astronaut exploration mission planned by the National Aeronautics and Space Agency (NASA) [1] as well as China's Chang'e-7 and Chang'e-8 missions [2]. There are numerous permanently shadowed regions (PSRs) in the LSP, where it is believed that there may be a large amount of water ice [3], [4], [5]. Water ice can serve as a vital resource for sustaining human life and in fuel production for future deep space exploration activities. The very rugged terrain and illumination conditions in the LSP make both robotic and human exploration activities challenging. Precise landing for exploration missions in the LSP is critical, as large landing errors may lead to probes falling into PSRs. The difficulty of landing navigation for lunar south polar exploration missions is much greater than that of previous missions in other mid to low latitude regions. Currently, an effective solution for safe landing in the LSP is to use high-resolution lunar topographic mapping products for optical navigation. Indeed, the upcoming missions in the LSP heavily relies on high-precision maps to fully characterize the landing sites and assist landing navigation [6], [7], [8], [9]. Although many lunar exploration missions have carried out detailed remote sensing observations of the Moon and published various mapping products, the geolocation accuracy, spatial resolution, and valid coverage of existing mapping products in the LSP still cannot meet the precise landing requirements for upcoming exploration missions. Specifically, submeter-level image mosaics and meter-level terrain models are very scarce for the LSP, but the demand for these is very high.

Mapping products such as digital elevation models (DEMs) of extraterrestrial bodies are mainly generated using optical images or laser altimetry data [10], [11], [12], [13], [14]. Optical images can be used to generate terrain models through stereo photogrammetry (SPG) [15] or shape from shading (SfS) methods [16]. The terrain models generated from optical images have the merits of higher spatial resolution, wide operating range, and low cost. The main drawback of optical imaging methods is that they are easily affected by illumination conditions and image texture. The height accuracy of the generated terrain models based on the SPG method is significantly influenced by the

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stereo intersection angles, and the quality of the generated terrain models will be greatly reduced when the image lacks texture information. In principle, the SfS method can generate DEMs approaching the pixel scale of the original images by utilizing the image pixels' brightness and a photometric function to solve the surface topography [17]. However, the entire process of the SfS method is very time-consuming and is usually more suitable for mapping local regions. Laser altimetry data can be used to derive large-scale terrain models through interpolation, with the main advantages of very high vertical accuracy and being irrelevant to illumination conditions. The geolocation accuracy and reconstruction quality of terrain models generated using laser altimetry data are affected by the spacecraft's timing, position, and attitude information. Moreover, laser altimetry data always exhibit large cross-track spacing, leading to the limited spatial resolution of these terrain models, especially in low-latitude areas of the planetary surface. In addition, a large volume of research has been carried out on the tie point extraction and three-dimensional reconstruction of planetary images based on deep learning methods [18], [19], [20], [21]. Although these deep learning methods have shown strong application prospects, there are still certain problems when directly applying them to support exploration missions, including obtaining sufficient training sample data and evaluating geolocation accuracy.

In this article, we propose a new image matching and control network generation method for the photogrammetric processing of lunar south pole images. Specifically, we use the existing low-resolution lunar terrain models, namely the lunar orbiter laser altimeter (LOLA) DEMs, to render shaded images, and then we conduct image matching between the shaded DEM images and the approximate orthophotos of lunar reconnaissance orbiter (LRO) narrow-angle camera (NAC) images. Then, the control network is automatically generated based on our accurate control measure's prediction method. Our method can greatly increase the success rate of image matching in the LSP, and improves the quality of the control network. Because the accuracy of the predicted control measures is relatively high, a small search window can be used to find the conjugate control measures. Consequently, the processing time of control network construction is also considerably reduced. The main contributions of our work are as follows.

- 1) We propose an image matching method based on the approximate orthophotos of LRO NAC images and shaded DEM maps, which can greatly improve the matching quality for stereo images of the LSP. This strategy can significantly reduce the adverse effects of large differences in illumination conditions during image matching.
- 2) We design an accurate refinement method for candidate control measures based on the matching results between the approximate orthophotos of LRO NAC images and the shaded DEM maps. This ensures that the start positions of the candidate control measures are very close to the exact positions, and a small search window size is enough to match sufficient and accurate feature points.
- 3) We provide an optimized control network generation process integrating the USGS ISIS's software modules and the proposed control measures' refinement method. This

can greatly improve the success rate of image matching and enhance the quality of the automatically generated control network for the lunar south pole images based on the existing high-resolution (e.g., 10 m/pixel) LOLA DEM.

II. RELATED WORK

The LRO NAC provides remote sensing images of the Moon with the highest spatial resolution (0.5–1.5 m/pixel) at present [22]. However, the LRO NAC is not designed to be a stereo mapping camera. Although the LRO NAC can obtain stereo observations by acquiring images from two orbits (with at least one off-nadir slew) [23], the LRO NAC slew opportunities are very limited. Therefore, it is impossible to use the SPG method to produce DEMs in large-scale areas based on LRO NAC images. The LOLA DEM is produced based on the data obtained from the laser altimeter instrument onboard the LRO. The average accuracy of the LOLA DEM is better than 20 m in the horizontal direction and about 1 m in radius direction after crossover correction [24], [25]. The LOLA and SELEnological and Engineering Explorer (SELENE) Kaguya teams provided an improved near-global lunar DEM product with a spatial resolution of 60 m/pixel [26], namely, SLDEM2015. Furthermore, terrain models with higher spatial resolution (e.g., 5–30 m/pixel) are available polewards of 75 °S. The LOLA DEM and SLDEM 2015 are usually used for height control in the photogrammetric processing of lunar images [27]. However, because the spatial resolution of the LOLA DEM and SLDEM 2015 is limited, it is difficult to use them as horizontal control for LRO NAC images. Recently, to better support the upcoming exploration missions in the LSP, multiple teams have produced local or regional mapping products for the LSP based on laser altimetry data obtained from the LOLA [28], [29], [30]. For example, Barker et al. [28] generated a new high-resolution terrain model for potential landing sites in the LSP by iteratively adjusting the LOLA tracks to the LOLA-based DEM in a self-consistent fashion. However, even if there are a large number of repeated laser altimetry tracks in the LSP, the spatial resolution of the interpolated DEMs can hardly exceed 5 m/pixel even in areas above 85 degrees south latitude, which is significantly lower than the terrain models generated by optical image methods.

China's Chang'e-2 mission has acquired global lunar images with a spatial resolution of 7 m/pixel [31], [32]. Based on the photogrammetric processing of the acquired stereo images, the generated mapping products (i.e., CE2TMap2015), including global lunar image mosaics with a spatial resolution of 7 m/pixel and global lunar DEM with a grid spacing of 20 m/pixel, were provided to the public. However, due to the limited observation period, the returned images near the PSR in the LSP were not obtained to the greatest extent. Consequently, both the geolocation accuracy and spatial resolution of the image mosaics and the terrain models in polar regions above 85 °S are relatively inferior to those of other areas. However, these areas are the main target areas for future lunar exploration activities. The terrain camera on board SELENE Kaguya spacecraft launched by the Japan Aerospace Exploration Agency (JAXA) acquired lunar images

with a spatial resolution of 10 m/pixel [33], [34]. Similarly, the coverage and the accuracy of the generated near-global mapping products in the LSP were also limited.

Compared with China's Chang'e-2 and JAXA'S SELENE missions, the LRO mission has a longer observation period and better coverage for the LSP. The LRO spacecraft was launched in 2009 and is still on-orbit to obtain scientific data [22]. A longer observation period is beneficial for capturing images covering more areas in the LSP, and images obtained at different times can complement each other, especially near the PSRs. There are more than 500 000 LRO NAC images within the range from 60° south latitude to the south pole. Therefore, mapping products derived from LRO NAC images show obvious advantages in supporting the engineering operations of the upcoming exploration missions in the LSP [35], [36]. For one thing, the generated sub-meter level image mosaics for local or regional areas based on the LRO NAC images play a vital role in selecting landing sites and planning the rover path [37]. Moreover, even if the SPG method is not suitable for generating large-scale DEMs based on LRO NAC images due to the extremely small stereo intersection angles, we can adopt the SfS technique to generate DEMs with spatial resolution approaching the pixel scale of the original images [38]. The SfS technique requires knowledge of the spacecraft's position and attitude to calculate the incidence angle, emission angle, and photometric function. Thus, a satisfying bundle adjustment solution is a prerequisite for both SPG and SfS methods.

The challenge in photogrammetric processing for LSP images is mainly due to the unique very rugged terrain and illumination environment in the LSP, which makes the photogrammetric processing of LRO NAC images extremely difficult [39]. Specifically, the images in the LSP contains a large amount of shadows. Moreover, changes in illumination conditions result in significant differences in the appearance of images. Consequently, it is difficult to construct a large-scale control network for LSP images. One of the most difficult steps in the complex photogrammetric processing flow of planetary remote sensing images is the construction of a high-quality control network, which mainly depends on automatic image matching. Unfortunately, the low Sun elevation angle, very rugged terrain, and rapidly changing lighting directions in the LSP result in that the stereo images obtained at different times show large differences in appearance. Consequently, image matching in the LSP always fails with conventional methods such as the scale invariant feature transform (SIFT) algorithm. It is noted that sparse image matching (mainly used for generating control networks) based on deep learning techniques such as SuperPoint and SuperGlue show significant advantages compared with traditional methods [40], [41], [42], but they will fail when the images have large differences in illumination conditions and less texture information.

Chen et al. [43] focused on the large-scale bundle adjustment of LRO NAC images for the LSP. The authors integrated a reference DEM constraint into bundle adjustment to solve the weak convergence geometry of LRO NAC images, and reduced the relative positioning discrepancies to less than 1 m. However, the researchers focused on outlier elimination in the tie points

caused by mismatches, and did not provide detailed methods on how to obtain sufficient high-quality tie points. Xue et al. [44] proposed an effective feature-matching scheme specifically for high-resolution planetary images. They designed an image partitioning method for overlapping areas, which can handle the complex polygonal overlapping regions and simplify the matching problem for large images. Wu et al. [45] proposed an illumination-invariant SIFT matching method for planetary images and obtained approximately 40% to 60% more matches than the classical SIFT method. Huang et al. [46] proposed an improved image matching method for high-resolution planetary images through extracting low-high frequency features by building double-frequency scale space. However, these methods are not specifically designed for images in the LSP, so the application effects in this context are unknown. Indeed, there are few studies focusing on image matching methods for images from the LSP. Liu et al. [47] designed an optimized image matching algorithm according to the characteristics of the gray histogram of images in the LSP to increase the image contrast and improve the matching quality. However, the matching results are still affected by changes in the illumination conditions.

The existing image matching methods provided in famous open-source planetary image processing software, i.e., the United States Geological Survey (USGS) Integrated Software for Imagers and Spectrometers (ISIS) [48] and the National Aeronautics and Space Administration (NASA) Ames Stereo Pipeline (ASP) [49], [50], do not perform well for images of the LSP. The default parameters and strategies in these software tools often fail to match feature points or result in a large number of mismatched points. Specifically, the control-network-related software tools in USGS ISIS such as "pointreg" and "findfeatures" cannot fully automate the generation of control networks for images of the LSP, typically requiring a significant amount of manual editing work. For NASA ASP, although multithreaded parallel and block processing optimization have been carried out for control network generation software tools, satisfactory results still cannot be obtained for images of the LSP due to the lack of targeted algorithm optimization.

III. METHODOLOGY

The key to constructing a high-quality control network is finding enough conjugate feature points for overlapping images. In this study, we propose a control network construction method based on image matching between the shaded DEM maps and the approximate orthophotos of the LRO NAC images for specifically mapping in the LSP. The shaded DEM maps are generated using the same photograph time of the corresponding on-orbit LRO NAC images, which ensures that the shaded DEM maps have very similar illumination conditions as the original LRO NAC images. The matching results between the shaded DEM maps and the LRO NAC DOMs can be utilized to accurately predict or refine the positions of the candidate control measures. Therefore, the use of shaded LOLA DEM maps can improve the control measures' accuracy, which can significantly improve the efficiency and accuracy of image matching. Consequently, it can enhance the quality of the generated control network.

A. Existing Control Network Generation Process Based on USGS ISIS

A high-quality control network is essential for a successful bundle adjustment. The most time-consuming step in the whole process of planetary photogrammetric processing is constructing and editing the control network. USGS ISIS is widely used to conduct the photogrammetric processing of planetary images. It provides many software tools for generating and editing control networks, including “autoseed,” “cnetref,” “pointreg,” “cneteditor,” “qnet,” and “qmos.” Here, we use the terms related to the control network consistent with USGS ISIS. Specifically, a control network is composed of multiple control points, which is used as inputs to refine the exterior orientation parameters of on-orbit planetary images via bundle adjustment. The control points used in USGS ISIS are similar to the tie points used in the photogrammetric processing in Earth observation fields. A control point is composed of multiple control measures obtained on multiple overlapping images. A control measure is a feature point identified on an image. All control measures on a control point should be relative to the same ground point. Usually, there is a reference control measure and several candidate control measures for a control point in the process of generating a control network. Specifically, the reference control measure is selected by some interest operator. It is a prominent feature point on an image, which facilitates the following matching operations. Then, the main work in generating control network is determining the exact locations of these candidate control measures through image matching methods.

Here, we briefly introduce the control network generation process based on USGS ISIS. It first generates the initial control network through the “autoseed” program, and then it invokes the “cnetref” program to determine the location of the reference control measures as well as other candidate control measures. The main purpose of this step is to identify a reference control measure with higher interest value for a control point, which will be more conducive to image matching. Note that at this stage, the locations of the candidate control measures deviate from the exact positions. Then, it invokes the “pointreg” program to conduct matching operations between the reference control measures and other candidate control measures for each control point. What we need to emphasize is the determination of the start positions of the candidate control measures has the greatest impact on the whole process of control network construction. On one hand, if the candidate control measures’ start positions are inaccurate, a large search window may be required and more time may be needed for matching. On the other hand, when the start positions of the candidate control measures are too far away from the real positions, there is a risk of false matching points.

Indeed, in terms of USGS ISIS, the start positions of the candidate control measures are determined using the spacecraft’s position and pointing information as well as a reference terrain model. Specifically, given the reference control measure of a control point, its corresponding ground point can be determined through ray intersection with the related terrain model (i.e., an existing DEM). Then, other candidate control measures’ pixel coordinates can be computed via back projection operations

from ground coordinates to pixel coordinates. Therefore, the accuracy of the spacecraft’s position and pointing information, as well as the reference DEM, determines the accuracy of the start positions of the candidate control measures.

The spacecraft’s position and pointing information are acquired from SPICE kernels. There are usually three precision levels of SPICE kernels: predicted level, reconstructed level, and smithed level. Predicted-level SPICE kernels have the lowest accuracy and are not used for mapping purposes. Smithed-level SPICE kernels have the highest accuracy, but the number of images involved is limited. Indeed, the reconstructed-level SPICE kernels are the main data used for photogrammetric processing. Due to the large amount of LRO NAC images (more than two million), there is a lack of detailed evaluation results on the prediction accuracy of candidate control measures based on the reconstructed-level SPICE kernels. However, according to our experiments and some related papers, the prediction accuracy can reach dozens of pixels for most stereopairs, and an error of more than 100 pixels may occur in poor conditions.

B. Optimized Control Network Generation Process Based on Refining Tie Points

The main feature of our method is the ability to provide accurate start positions of the candidate control measures by image matching between the LRO NAC DOMs and the shaded DEM maps. Indeed, we do not rely on the accuracy of SPICE kernels to determine the start positions of the candidate control measures. The accuracy and resolution of available LOLA DEMs affects the prediction accuracy of the candidate control measures for our method. In order to make full use of the various software tools provided by USGS ISIS, we integrate our control measures’ refinement method with the processing flow of the control network generation in USGS ISIS. The optimized control network generation process based on our method is shown in Fig. 1.

The detailed steps are as follows.

- 1) Prepare input data, including a LOLA DEM with appropriate resolution, LRO NAC images, and SPICE kernels.
- 2) Conduct LRO NAC image preprocessing using USGS ISIS’s programs, including data format conversion using “Ironac2isis,” radiometric calibration using “Ironacecho” and “Ironacal,” and SPICE relation using “spiceinit.”
- 3) Generate approximate orthophotos of LRO NAC images using ISIS’s “cam2map” program based on the initial exterior orientation parameters and reference DEM, and downsample them to the same resolution as the reference DEM.
- 4) Clip a DEM according to the LRO NAC DOM’s geographic coordinate range, and then generate a shaded DEM map using the photograph time of the on-orbit LRO NAC image.
- 5) Conduct image matching between the LRO NAC DOMs and the corresponding shaded DEM maps based on the SIFT feature matching method and pyramid matching strategy. Then, construct the homograph transformation matrix for all stereopairs of LRO NAC DOMs.

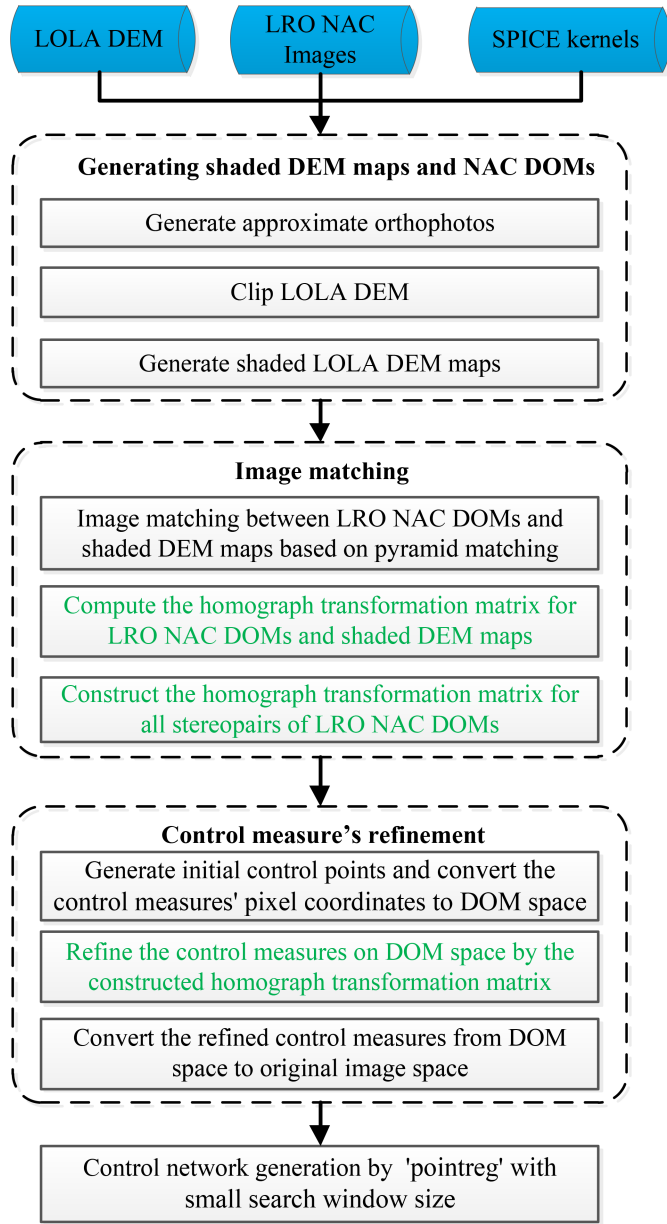


Fig. 1. Control network generation process based on our method. The green font represents the characteristic steps based on our method.

- 6) Generate initial control network by using ISIS's "autoseed" program. Then, for a reference control measure on a control point, compute its pixel coordinates in the DOM space and then predict the pixel coordinates of the candidate control measures in the orthophoto space using the constructed homograph transformation matrix.
- 7) Convert the predicted control measures' pixel coordinates from the orthophoto space to the original image space.
- 8) Generate the control network using the normalized correlation coefficient (NCC) method with an appropriate search window size.

The LRO NAC images are acquired by pushbroom imaging camera. Thus, the original LRO NAC images have significant distortions. The success rate of matching directly on the

original LRO NAC images is relatively low. Using the strategy of matching on approximate orthophotos in the process of control network construction has the merits of improving the success rate of matching. The shaded LOLA DEM map can provide a reference DOM with certain geometric accuracy, and it is consistent in appearance with the on-orbit NAC image. The shaded LOLA DEM map based on high-resolution topography products (e.g., 5–30 m/pixel LOLA DEM) often has higher accuracy than the initial positioning accuracy provided by SPICE kernels. Since we can use homograph transformation to establish the coordinate transformation relationship between each LRO NAC DOM and the corresponding shaded LOLA DEM map, we do not directly match two LRO NAC DOMs. As an alternative, we can establish a coordinate transformation relationship between any two NAC DOMs by using shaded LOLA DEM maps as a bridge. This helps to solve the very difficult problem of matching lunar south polar images. The pixel coordinates of feature points obtained by matching on the approximate orthophotos can be converted to the original image space based on the principle of back projection of linear pushbroom images, and then they can be used as image observations for bundle adjustment.

The search window size used in the NCC method is related to the spatial resolution of the reference DEM. Specifically, when the available LOLA DEMs have a relatively high resolution such as 10 m/pixel, the prediction accuracy based on our method is relatively high, and a small search window size is enough for matching. Since we transform various coordinates in the DOM space, original image space, geodetic coordinates, and map projection coordinates, we should ensure that all DOMs and shaded DEM maps have the same map projection information. In addition, the spacecraft's position and pointing information should remain unchanged during the process of refining the control measures. This is because bundle adjustment changes the exterior orientation parameters of LRO NAC images, thus affecting the accuracy of the coordinate transformation between the DOM space and original image space.

C. Generation of Shaded DEM Maps Using the Photograph Time of On-Orbit LRO NAC Images

Generally, shaded DEM maps are used to visualize the DEM as they contain enough information to allow the observer to recover the terrain relief [51]. Based on this idea, we can also use shaded DEM maps as reference images for image matching with on-orbit images. Furthermore, if shaded DEM maps are produced with the same illumination conditions as on-orbit images, they will have a very similar appearance. Consequently, image matching between on-orbit images and shaded DEM maps will be much easier. Shaded DEM maps can be generated using the following equation:

$$\text{Shade} = \frac{1 + p_0 \times p + q_0 \times q}{\sqrt{1 + p \cdot p + q \cdot q} \times \sqrt{1 + p_0 \cdot p_0 + q_0 \cdot q_0}} \quad (1)$$

where p and q are the two orthogonal gradients of the DEM data, and p_0 and q_0 can be computed with the Sun azimuth angle and

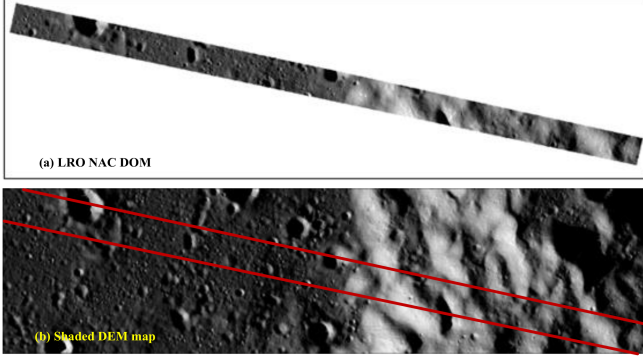


Fig. 2. LRO NAC DOM and the corresponding shaded DEM map. (a) Top figure is the DOM of M153386201RE. (b) Bottom figure is the corresponding shaded DEM map, which was rendered based on the LOLA DEM with a spatial resolution of 30 m/pixel. The red solid line in the shaded DEM map marks the corresponding geographical range of the LRO NAC DOM. The LRO NAC DOM was generated using polar stereographic map projection.

the solar elevation angle by using the following equation:

$$\begin{cases} p_0 = -\cos(a) \times \tan(e) \\ q_0 = -\sin(a) \times \tan(e) \end{cases} \quad (2)$$

where a and e are the Sun azimuth angle and solar elevation angle expressed in radians, respectively. For the solar azimuth angle, its value range is 0–360 decimal degrees with 0° at three o'clock, increasing in the clockwise direction. The Sun azimuth angle and the solar elevation angle can be determined based on Spacecraft Planetary Instrument C-matrix Events (SPICE) kernels [52]. Specifically, given a specific time T , the Sun's position relative to the Moon in a body-fixed coordinate frame can be calculated using the corresponding SPICE kernels. Then, given a ground point $P(X, Y, Z)$ on the lunar surface and the corresponding DEM, the normal of the ground point P and the vector from the ground point to the Sun can be calculated. Next, the Sun azimuth and the solar elevation angles are derived, and the gray value for each pixel on the shaded DEM map can be calculated based on (1). Thus, if we use the same photograph time of the on-orbit LRO NAC images to generate the shaded DEM maps, they will have the same Sun azimuth angle and solar elevation angle. Consequently, their appearance will be very similar.

Fig. 2 is an example of a shaded DEM map using the LOLA DEM with a spatial resolution of 30 m/pixel. As can be seen, the shaded DEM map exhibits an appearance very similar to the DOM of the corresponding on-orbit LRO NAC image. It is noted that the size and direction of the shadows in the two figures are almost the same. Obviously, image matching between them is relatively easy. The detailed steps of generating the shaded DEM maps corresponding to LRO NAC images are as follows.

- 1) Generate the approximate orthophotos of LRO NAC images based on SPICE kernels and a reference terrain model (i.e., available LOLA DEM). Downsample the DOMs to the same spatial resolution as the LOLA DEM.
- 2) Compute the geographic coordinate range of the LRO NAC DOM, and clip the same range of DEM chip from the available LOLA DEM.

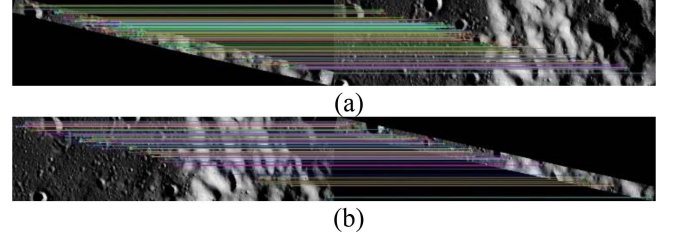


Fig. 3. Image matching results between LRO NAC DOMs and shaded DEM maps. The left part in the top figure and the right part in the bottom figure are the DOMs of M153379413LE and M153386201RE, respectively. The corresponding shaded DEM maps are rendered based on a LOLA DEM with a spatial resolution of 30 m/pixel.

- 3) According to the photograph time of the on-orbit LRO NAC image, compute the Sun's position as well as the solar azimuth angle and the solar elevation angle for each pixel on the clipped DEM. Then, compute the shade value for each pixel on the clipped DEM based on (1).
- 4) Using the computed shade value as a grayscale value, generate the shaded DEM map for each LRO NAC DOM.

D. Image Matching Between Shaded DEM Maps and Approximate Orthophotos of LRO NAC Images

Since the shaded DEM maps have a very similar appearance to the corresponding LRO NAC DOMs, matching them becomes much easier. Based on this consideration, we used a mature matching method in photogrammetry and computer vision community, namely, the SIFT algorithm, for matching the shaded DEM maps and the LRO NAC DOMs. Due to the high resource consumption of the SIFT matching method, a pyramid matching strategy and a block processing approach are adopted when processing large images. Furthermore, there are large differences in the spatial resolution between the LRO NAC images and the available LOLA DEM products. This may reduce the number of matched feature points to some extent. As mentioned earlier, the spatial resolution of the available LOLA DEM products varies from 5 to 60 m/pixel in the LSP. Moreover, the higher the latitude, the higher the resolution of the LOLA DEM. As can be seen in Fig. 3, although the resolution of the LOLA DEM product is much lower than that of the LRO NAC images, enough feature points can be matched using the classical SIFT feature matching method. This is mainly due to the fact that the same illumination conditions lead to a very similar appearance between the shaded DEM maps and the corresponding LRO NAC DOMs. As a comparison, when we directly match two LRO NAC images or their DOMs, neither the traditional feature matching methods (e.g., SIFT) nor the deep learning matching methods (e.g., SuperPoint) can obtain satisfactory results.

Although we do not conduct image matching directly between two LRO NAC images or their DOMs, we can construct the coordinate transformation relationship for them by using the matching results obtained from shaded DEM maps as a transition. According to our comprehensive experiments, the number of matched feature points between LRO NAC DOMs and the corresponding shaded DEM maps meets the needs of establishing a coordinate transformation relationship for

stereopairs of LRO NAC images. Even in areas above 85° south latitude, hundreds of feature points can be matched based on our method by using a LOLA DEM with 10m/pixel (see Section IV). In the low-latitude area where only a LOLA DEM with 60 m/pixel is available, dozens of feature points can be matched. Several evenly distributed conjugate feature points are sufficient to construct a homograph transformation matrix for two images. It is noted that the closer to the LSP, the more difficult it is to match the images. However, we should also know that the closer to the LSP, the higher the resolution of the LOLA DEM currently available. Therefore, the available LOLA DEM products ensure that our method has good applicability in the photogrammetric processing of lunar south pole images.

The strategy of image matching in approximate orthophoto space has been widely used in the planetary mapping community [53], [54] due to the absence of moving objects (e.g., cars) or artificial buildings as well as better continuity on planetary surfaces. The idea of image matching in orthophoto space has the advantages of eliminating geometric distortions caused by both terrain undulations and oblique photography as well as making the spatial resolution of stereo images consistent. This practical strategy was also integrated into the photogrammetric processing flow in NASA ASP [49], [50], which was used for both automatic tie point extraction and DEM generation. Because we can obtain matched feature points between LRO NAC DOMs and the corresponding shaded DEM images in orthophoto space, the coordinate transformation relationship between any LRO NAC DOM and its corresponding shaded DEM image can be constructed by constructing a homograph transformation matrix using the matched feature points. Then, the corresponding homograph transformation matrix for two LRO NAC DOMs can be computed, and the coordinate transformation relation between them can be derived. Obviously, it is impossible to establish a stable coordinate transformation relationship based on a homograph transformation matrix in the original image space for a stereopair of linear pushbroom images. However, it is feasible in orthophoto space due to the fact that the image distortions caused by pushbroom imaging has been removed to the greatest extent.

E. Various Coordinate Transformation Flow for Refining Control Measures

Since we first refine the candidate control measures in the orthophoto space and then convert them to the original image space, this process involves a series of coordinate transformations based on the camera model of the LRO NAC images.

The coordinate transformation process from the orthophoto space to the original image space for a control measure is as follows.

- 1) Convert the pixel coordinates $(S_{\text{dom}}, L_{\text{dom}})$ of a control measure in the DOM (i.e., orthophoto) space to map projection coordinates $(\text{proj}X, \text{proj}Y)$ using the geographic coordinate information related to the DOM.
- 2) Convert the map projection coordinates $(\text{proj}X, \text{proj}Y)$ to geodetic coordinates $(\text{Long}, \text{Lat})$ using the corresponding inverse map projection transformation.

- 3) Based on the reference DEM used in the DOM generation step, determine the corresponding height H using the map projection coordinates $(\text{proj}X, \text{proj}Y)$. Then, convert the geodetic coordinates $(\text{Long}, \text{Lat}, H)$ to space rectangular coordinates (X, Y, Z) .
- 4) Compute the pixel coordinates $(S_{\text{ori}}, L_{\text{ori}})$ in the original image space according to the ground point coordinates (X, Y, Z) using the back projection algorithm of LRO NAC images.

The coordinate transformation process from the original image space to the DOM space for a control measure is also used in our method. The detailed steps are as follows.

- 1) Given the pixel coordinates $(S_{\text{ori}}, L_{\text{ori}})$ of a control measure in the original image space and the reference DEM, the ground point coordinates (X, Y, Z) can be computed via ray intersection with the reference DEM.
- 2) Convert the ground point coordinates (X, Y, Z) to geodetic coordinates $(\text{Long}, \text{Lat}, R)$.
- 3) Convert the geodetic coordinates $(\text{Long}, \text{Lat})$ to map projection coordinates $(\text{proj}X, \text{proj}Y)$ using the forward map projection transformation.
- 4) Convert the map projection coordinates $(\text{proj}X, \text{proj}Y)$ to pixel coordinates $(S_{\text{dom}}, L_{\text{dom}})$ in the DOM space.

Because the LRO NAC acquires images via the pushbroom imaging principle, each scan line has its own exterior orientation parameters. The back projection coordinate transformation (i.e., ground-to-image) requires iterative computations [55]. According to the matched feature points between LRO NAC DOMs and the corresponding shaded DEM maps, the coordinate transformation based on homograph transformation for any two LRO NAC DOMs can be indirectly obtained by using the shaded DEM maps as a bridge. Then, given a pixel coordinate on an LRO NAC DOM, the conjugate feature point on other overlapping LRO NAC DOMs can be determined based on the constructed homograph transformation matrix. Thus, the candidate control measures' start positions on a control point can be refined based on this idea. As shown in Fig. 4, the process of refining candidate control measures is as follows:

- 1) Construct the homograph transformation matrix for a stereopair composed of two LRO NAC DOMs.
- 2) For a control point, convert its reference control measure's pixel coordinates from original image space to orthophoto space.
- 3) According to the pixel coordinates of the reference control measure on an LRO NAC DOM, the pixel coordinates of the candidate control measures on the other LRO NAC DOMs can be computed based on the constructed homograph transformation matrix.
- 4) Convert the pixel coordinates of the predicted candidate control measures from the orthophoto space to the original image space.
- 5) Perform steps 2 to 4 for all control points.

IV. EXPERIMENTS AND RESULTS

A. Test Datasets

The proposed method was implemented based on USGS ISIS and OpenCV. Specifically, the refinement of the candidate

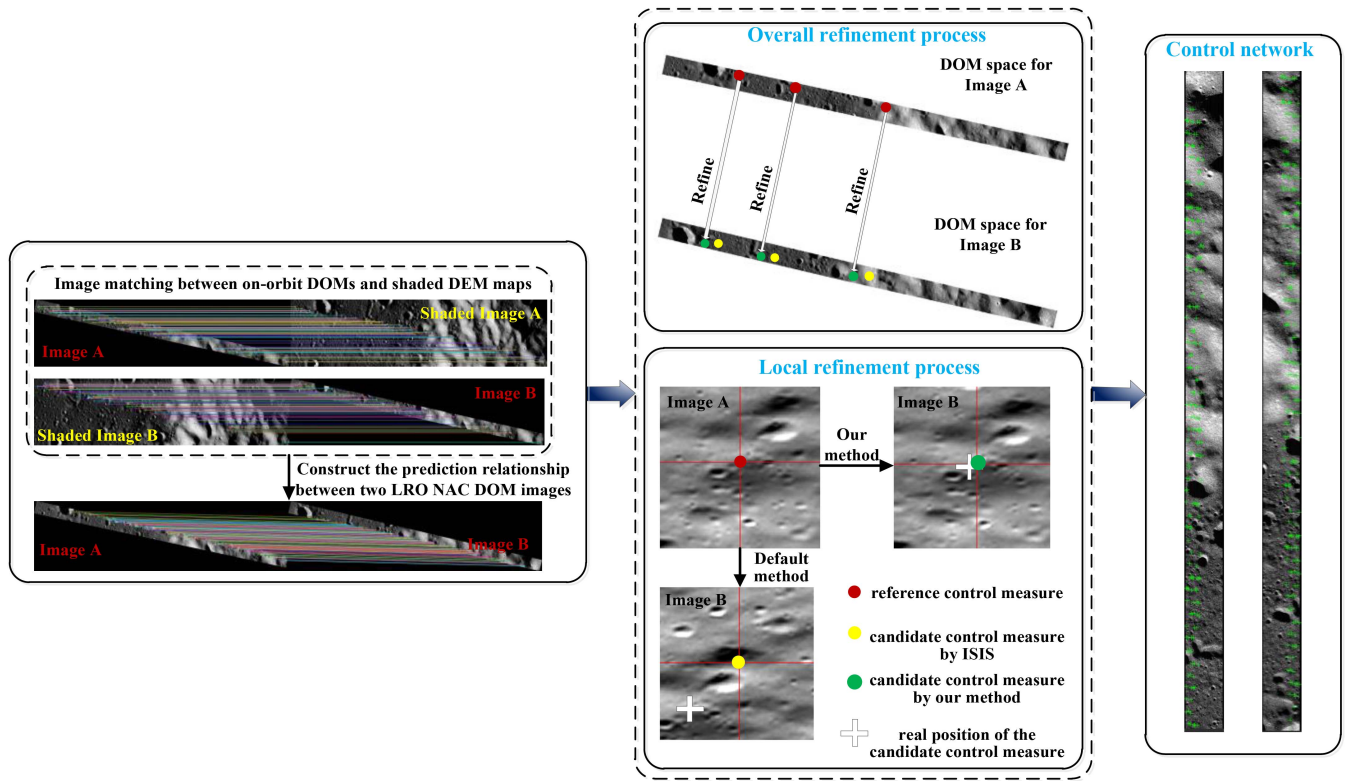


Fig. 4. Illustration of the refinement of candidate control measures based on our method. In the middle part of the figure, the red dots on image A indicate the reference control measures; the green and yellow dots on image B indicate the candidate control measure predicted by our method and USGS ISIS's method, respectively.

control measures' start positions was developed using the control network module of USGS ISIS. Image matching between the shaded DEM images and LRO NAC DOMs was developed based on OpenCV. The independently developed software modules for image matching and control network refinements were developed using Qt 5.12.9 in the Ubuntu 20.04 operating system. ISIS 8.0.0 was used to perform the photogrammetric processing of LRO NAC images, including image preprocessing, bundle adjustment, and DOM generation. The hardware configurations were an Intel Core i7-7500U with a 2.70 GHz CPU and 16 GB RAM. The radius of the reference sphere for the Moon is 1737.4 km. We used polar stereographic map projection for DOM generation and pixel coordinate transformation between the orthophoto space and original image space. The test images include two sets of LRO NAC images located at the LSP, as listed in Fig. 5 and Table I.

The range of dataset 1 is 87.5° south latitude to 90° south latitude, and the range of Dataset 2 is 79.5° south latitude to 81.5° south latitude. The illumination conditions for Dataset 1 were worse than that of Dataset 2. NASA has provided optimized LOLA DEM products for different resolutions (from 5 to 60 m/pixel) in the LSP by iteratively co-adjusting the LOLA tracks in a self-consistent fashion [28]. The mapping product quality of the optimized LOLA DEM has been significantly improved, eliminating spurious noise points in the previous versions of LOLA DEM. Although the optimized LOLA DEM provides 5 m/pixel map products in the LSP, its coverage is limited. Thus, we did not conduct experiments using LOLA

DEMs with 5 m/pixel. The corresponding coverage of the used optimized LOLA DEMs (10, 30, and 60 m/pixel) is shown in Fig. 6. The 10, 30, and 60 m/pixel optimized LOLA DEMs can cover areas above 83°, 75°, and 60° south latitude, respectively. According to the available LOLA DEM product for the test data, we conducted five sets of experiments. The first three sets of experiments utilized Dataset 1, using LOLA DEMs with resolutions of 10, 30, and 60 m/pixel, respectively. The last two sets of experiments utilized Dataset 2, using LOLA DEMs with resolutions of 30 and 60 m/pixel, respectively. LOLA DEMs with different spatial resolutions can further verify the proposed method's feasibility.

B. Image Matching Results

The parameters listed in Table II were used for image matching. For Test 1, the LOLA DEM has a higher resolution of 10 m/pixel, and the data volume of the corresponding shaded DEM images and LRO NAC DOMs is approximately several hundred megabytes. Therefore, we initially used the third-level image pyramid overviews for image matching. For Test 2 and Test 4, we initially used the first-level image pyramid overviews for matching. For Test 3 and Test 5, since the spatial resolution of the LOLA DEM is 60 m/pixel and the amount of data for the shaded DEM maps and the corresponding LRO NAC DOMs is relatively small, we directly conducted image matching on the original image resolution. The image matching results from the image pyramids were only used as guidance information for

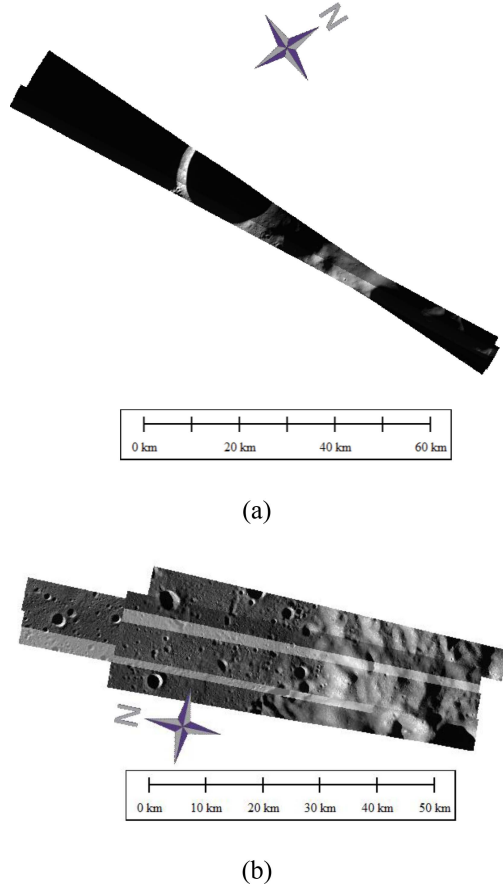


Fig. 5. Geographical distribution of the test images: (a) Dataset 1; (b) Dataset 2.

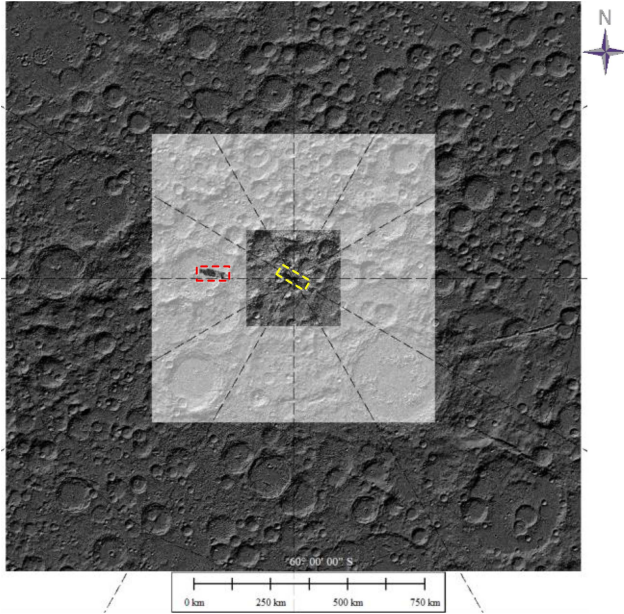


Fig. 6. Mapping products' coverage for LOLA DEMs with resolutions of 10, 30, and 60 m/pixel, respectively. The largest range with the darkest color is the 60 m/pixel LOLA DEM, the smallest range is the 10 m/pixel LOLA DEM, and the middle range is the 30 m/pixel LOLA DEM. The yellow dashed rectangle indicates Dataset 1. The red dashed rectangle indicates Dataset 2. The shaded DEMs were drawn using polar stereographic map projection.

TABLE I
BASIC INFORMATION OF THE TEST IMAGES

	Images	GSD (m)	Image size (lines $\times$ samples)	Incidence angle (degrees)	Emission angle (degrees)
Dataset 1	M110779506LE	0.82	52224 $\times$ 2532	87.75	1.70
	M110779506RE	0.82	52224 $\times$ 2532	87.78	1.16
	M110799874LE	0.81	52224 $\times$ 2532	87.73	1.71
	M110799874RE	0.81	52224 $\times$ 2532	87.76	1.16
	M110820244LE	0.80	52224 $\times$ 2532	87.75	1.71
	M110820244RE	0.81	52224 $\times$ 2532	87.78	1.16
Dataset 2	M153379413LE	0.87	52224 $\times$ 2532	83.57	1.71
	M153379413RE	0.87	52224 $\times$ 2532	83.61	1.17
	M153386201LE	0.88	52224 $\times$ 2532	83.45	1.71
	M153386201RE	0.88	52224 $\times$ 2532	83.49	1.17
	M153392986LE	0.89	52224 $\times$ 2532	83.43	1.71
	M153392986RE	0.88	52224 $\times$ 2532	83.47	1.17
	M153399771LE	0.89	52224 $\times$ 2532	83.40	1.71
	M153399771RE	0.89	52224 $\times$ 2532	83.44	1.17
	M158101667LE	0.88	52224 $\times$ 2532	83.91	1.71
	M158101667RE	0.88	52224 $\times$ 2532	83.96	1.17
	M158108448LE	0.89	52224 $\times$ 2532	83.92	1.71
	M158108448RE	0.89	52224 $\times$ 2532	83.96	1.17

Note: GSD Refers to ground sample distance.

block matching on the original resolution image. The block size of image matching is 512×512 , and the expected number of the extracted features in each block is 1000. The contrast threshold is 0.02, which is slightly lower than the default value of 0.04. This is because the images in the LSP always have poor contrast. Thus, a lower threshold helps extract more feature points, but will also result in more outliers in the matched feature points. In addition, taking into account both the number of matched points and the matching accuracy, we used different Lowe's ratio test thresholds for different experiments. For Test 3 and Test 5, because the LOLA DEM with a grid spacing of 60 m/pixel is used, the number of extracted feature points on the shaded DEM images and the corresponding LRO NAC DOMs are relatively small. To ensure sufficient matched feature points for control measure refinement, we used a higher Lowe's ratio test threshold value of 0.8. Indeed, a higher Lowe's ratio test threshold value will result in more outliers in the matching results. In order to solve the problem of more potential outliers caused by a lower contrast threshold and higher Lowe's ratio test threshold, we adopted a bidirectional strategy to eliminate outliers based on the RANSAC (Random Sample Consensus) method. The numbers of matched points between LRO NAC DOMs and the shaded DEM maps for Datasets 1 and 2 are presented in Figs. 7 and 8, respectively.

TABLE II
IMAGE MATCHING PARAMETERS

Tests	Datasets	Grid spacing	SIFT matching parameters			
			Overview level	Contrast threshold	Features in each chip	Lowe's ratio test threshold
Test1	Dataset 1	10m/pixel	3	0.02	1000	0.6
Test2	Dataset 1	30m/pixel	1	0.02	1000	0.6
Test3	Dataset 1	60m/pixel	0	0.02	1000	0.8
Test4	Dataset 2	30m/pixel	1	0.02	1000	0.6
Test5	Dataset 2	60m/pixel	0	0.02	1000	0.8

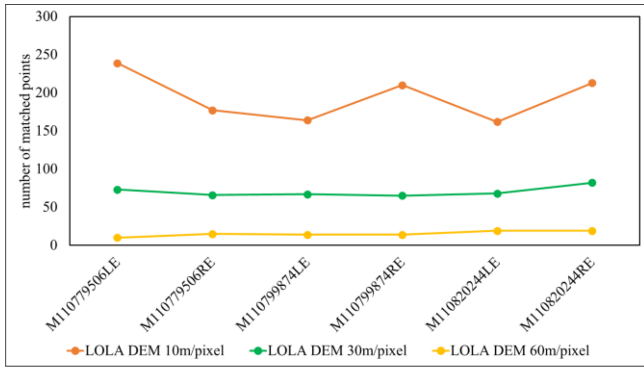


Fig. 7. Number of matched points between LRO NAC DOMs and the shaded DEM maps for Dataset 1.

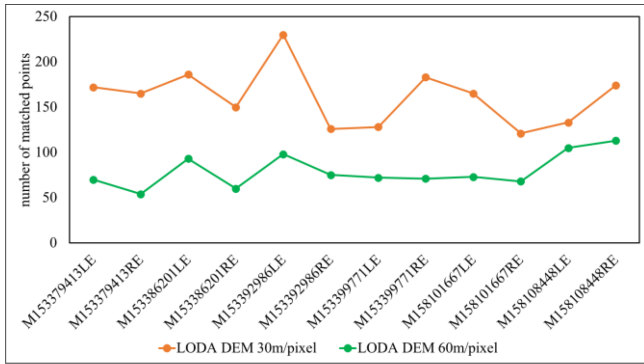


Fig. 8. Number of matched points between LRO NAC DOMs and the shaded DEM images for Dataset 2.

Taking M110779506LE in Dataset 1 and M153379413LE in Dataset 2 as example, we visualized the image matching effect between LRO NAC DOMs and the corresponding shaded DEM images by connecting the matched feature points into a straight line, and the results are shown in Fig. 9. As can be seen, for LOLA DEMs with the same resolution, the number of matching feature points in Dataset 2 is significantly higher than that in Dataset 1. This is mainly because the images in Dataset 1 have a higher latitude and poorer illumination conditions, as explained by the incidence angle of the test images listed in Table I. The experiments with the fewest matching feature points were obtained in Test 3, namely, the LOLA DEM with a grid spacing of 60 m/pixel for Dataset 1. Nevertheless,

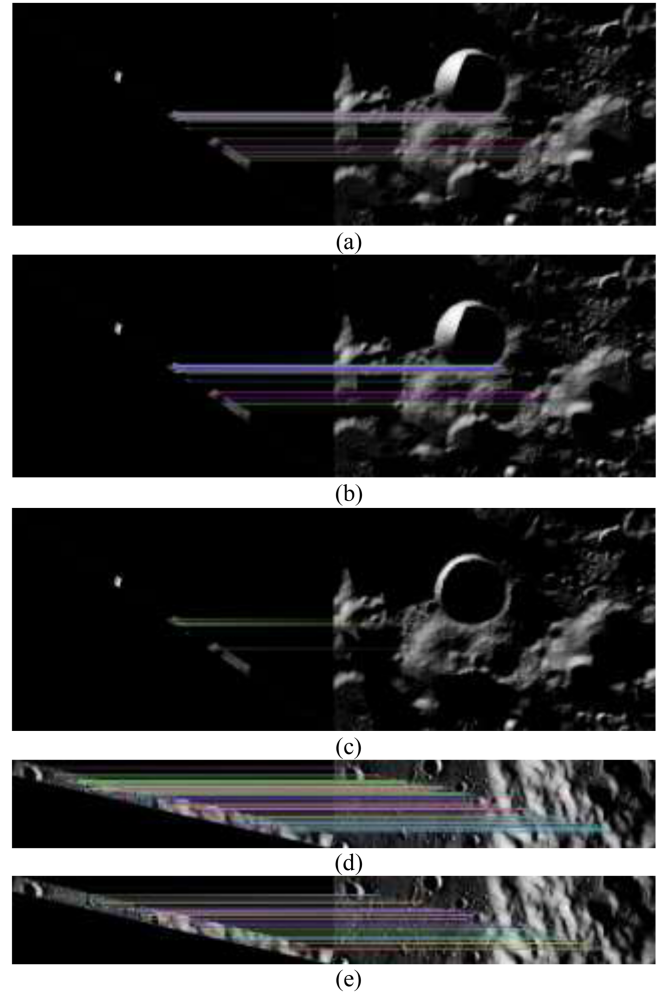


Fig. 9. Image matching results between LRO NAC DOMs and shaded DEM images with LOLA DEMs of different resolution. (a)–(c) are image matching results between LRO NAC DOMs and shaded DEM images for M110779506LE using LOLA DEM with 10 m/pixel (test1), 30 m/pixel (test2), and 60 m/pixel (test3), respectively. (d) and (e) are image matching results between LRO NAC DOMs and shaded DEM images for M153379413LE using LOLA DEM with 30 m/pixel (test 4) and 60 m/pixel (Test 5), respectively.

it still meets the minimum requirements for constructing the coordinate transformation matrix to refine control measures, and the distribution of the matched feature points is basically uniform. Note that for images with higher latitudes and closer proximity to the south pole, the available LOLA DEMs have

TABLE III
PREDICTION ACCURACY EVALUATION RESULTS BASED ON HOMOGRAPH TRANSFORMATION MATRIX FOR M110779506LE

grid spacing	Max. (pixels/meters)		Ave. (pixels/meters)		Std. (pixels/meters)	
	Sample	Line	Sample	Line	Sample	Line
10 m/pixel	2.24/22.4	1.87/18.7	0.0001/0.001	0.0001/0.001	0.67/6.7	0.58/5.8
30 m/pixel	1.70/51.0	1.51/45.3	0.0001/0.001	0.0001/0.001	0.53/15.9	0.48/14.4
60 m/pixel	0.48/28.6	0.67/25.2	0.0001/0.001	0.0001/0.001	0.22/13.3	0.29/17.5

Note: Max. indicates the maximum residual value between the predicted pixel coordinates and the actual pixel coordinates. Ave. indicates the average residual value between the predicted pixel coordinates and the actual pixel coordinates. Std. indicates the standard deviation residual value between the predicted pixel coordinates and the actual pixel coordinates.

higher spatial resolution. From the image matching results of Test 2 and Test 3, it can be seen that although there are a large number of shadows on the LRO NAC images, our method still obtained sufficient matching feature points. The main reason for this is that we used the same acquisition time of the on-orbit LRO NAC images when rendering the shaded DEM images, ensuring that the rendered shaded DEM images and the LRO NAC images had basically the same illumination conditions. Consequently, the image matching between the LRO NAC DOMs and the shaded DEM maps will generate more matched feature points. In comparison, SIFT matching between the LRO NAC images always failed for Dataset 1 and generated many incorrectly matched feature points for Dataset 2.

From the matching results of Tests 4 and 5, it can be seen that there are sufficient matched feature points even if the available LOLA DEMs have low resolution. This is mainly due to the fact that the test images in Test 4 and Test 5 have relatively better illumination conditions than those in Test 1, Test 2, and Test 3. Based on the analysis of the above results, it can be concluded that our method can be applied to LRO NAC images below 85°S latitude with lower resolution LOLA DEMs (60 m/pixel), and images above 85°S latitude with higher resolution LOLA DEMs (10 and 30 m/pixel). This happens to be consistent with the resolution of available LOLA DEMs of the LSP. Therefore, our method has strong applicability in the photogrammetric processing of remote sensing images of the LSP.

C. Prediction Accuracy Analysis

The prediction accuracy of the proposed method is essential to the refinement of the control measures. To evaluate the prediction accuracy, we conducted a feature point prediction experiment. First, we used the SIFT algorithm to acquire matched feature points between the DOM of M110779506LE and the corresponding shaded DEM image. Then, the homograph transformation matrix from the DOM of M110779506LE to the shaded DEM map could be derived. Next, using the feature points of the DOM of M110779506LE as input, the corresponding predicted feature points on the shaded DEM map could be computed. Then, we compared the differences between the pixel coordinates of the matched feature points and the pixel coordinates of the predicted feature points. The prediction accuracy results are shown in Fig. 10 and Table III.

D. Control Network Quality Analysis

We used USGS ISIS's "autoseed" program to generate initial candidate control points, and then the candidate control

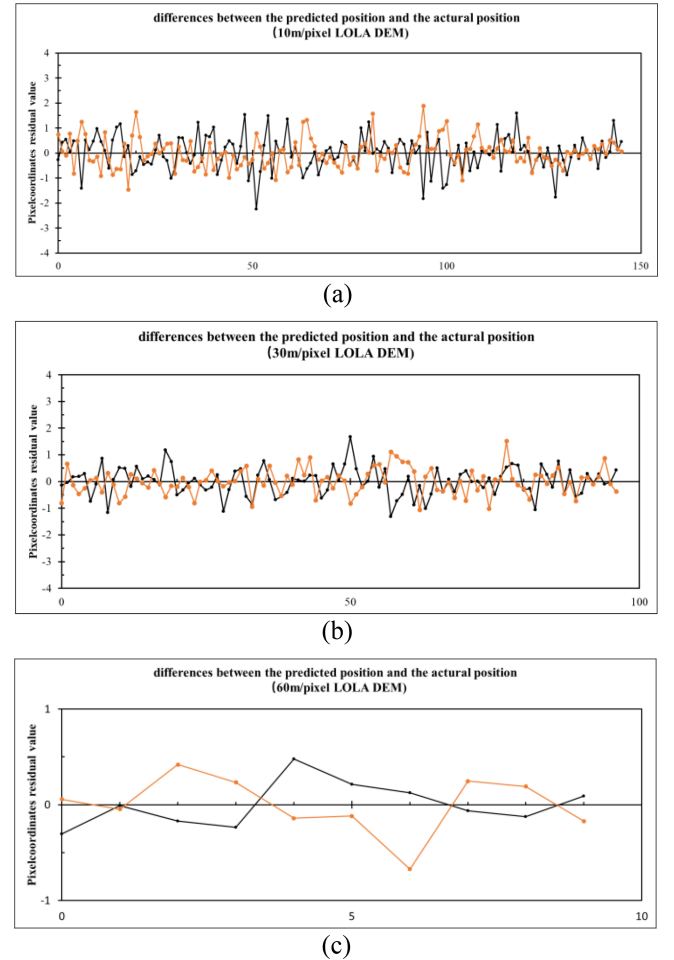


Fig. 10. Prediction accuracy evaluation results based on homograph transformation matrix for M110779506LE.

measures' pixel coordinates were generated based on USGS ISIS's "cnetref" program. Since image matching in the LSP is challenging, a dense control point layout scheme with 100 m in the X direction and 1000 m in the Y direction was used for initial control network generation. Consequently, the number of valid candidate control measures for Datasets 1 and 2 was 2024 and 5578, respectively. To visualize the distribution of control points, we drew the initially generated control points using ISIS's "qmos" program. The results are shown in Fig. 11.

Then, we used the proposed method to further refine the candidate control measures' pixel coordinates. The control network refined using our method was converted to ISIS's PVL control network format, and then we used ISIS's "cnetpv12bin" program

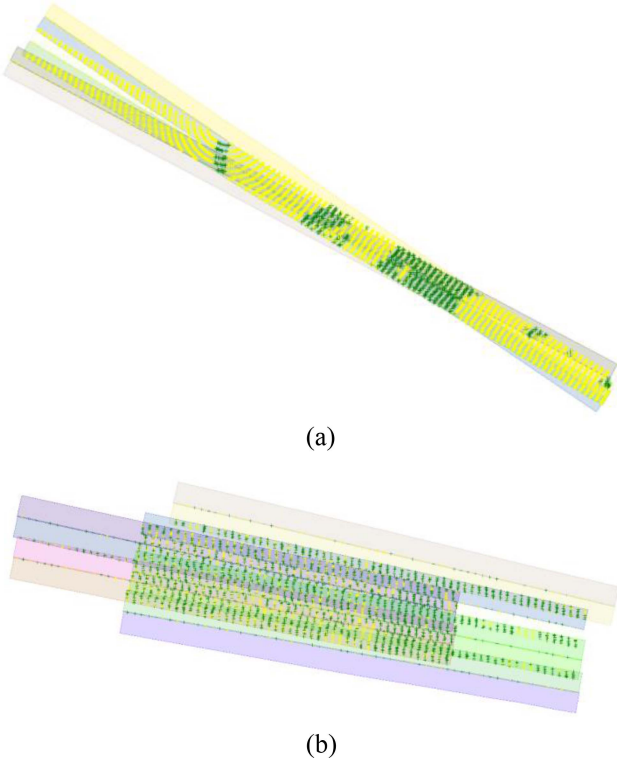


Fig. 11. Initial control point distribution for two datasets. Green cross indicates valid control points, and yellow cross indicates ignored control points. The ignored control points are always in the shadow area. (a) Distribution of initial control points for Dataset 1. (b) Distribution of initial control points for Dataset 2.

to derive a binary control network file that can be directly used by USGS ISIS's control point matching program "pointreg." The "pointreg" program adopts the NCC method to match conjugate points. The matching (or pattern) window size was specified as 19×19 . To better verify the prediction accuracy of the proposed method, different search window sizes (i.e., 31×31 , 51×51 , 71×71) were used. The numbers of matched control measures obtained using our method are listed in Fig. 12 and Table IV. For these tests, we used reconstructed-level SPICE kernels, but the accuracy of the SPICE kernels did not affect the prediction accuracy of our method. For comparison, we also conducted experiments using the control measures refined by ISIS's default processing flow (i.e., "cnetref"). For a more detailed accuracy comparison, we used both reconstructed-level and smithed-level SPICE kernels to refine the control measures derived from ISIS's "cnetref" program. The results are shown in Fig. 13 and Table V. Since different SPICE kernels were used, the number of initial candidate control measures also differed slightly.

A search window of 31×31 means that the pixel coordinates' deviation between the refined position and the exact position of the candidate control measure should be less than $(31-19)/2 = 6$ pixels. To search for conjugate control measures in such a small range, the accuracy of the refined control measures must be high enough. As can be seen in Tables IV and V, for Test 1 with a search window size of 31×31 , there were 1445 and 527 matched control measures obtained for our method and ISIS's

default processing flow, respectively. Note that these results were obtained based on the reconstructed-level SPICE kernels. This indicates that our method can provide a more accurate position for refined control measures with the adoption of the higher-resolution LOLA DEM (10 m/pixel). This accuracy is significantly better than the prediction accuracy obtained from reconstructed-level SPICE kernels. It is also noted that there were 2044 matched control measures obtained by ISIS's default processing flow for Test 1 with a search window size of 31×31 based on smithed-level SPICE kernels. This is relatively easy to understand, since smithed SPICE kernels are usually generated through bundle adjustment and can provide very accurate predicted positions for conjugate measures.

However, the available smithed SPICE kernels are very limited. Furthermore, even if all of the test images have smithed-level SPICE kernels, we should carefully verify the real effects on overlapping images. The smithed-level SPICE kernels of the involved LRO NAC images may not be obtained through an overall adjustment, which may lead to greater discrepancies for the adjacent LRO NAC images that have been processed using different bundle adjustments. This can be observed for Dataset 2. All test images in Dataset 2 have smithed-level SPICE kernels. As shown in Table V, for Dataset 2 with a 31×31 search window size, the number of matched control measures based on smithed-level SPICE kernels is more than the number of matched control measures based on reconstructed-level SPICE kernels. However, for Dataset 2 with a 51×51 and 71×71 search window size, the number of matched control measures based on smithed-level SPICE kernels is obviously less than the number of matched control measures based on reconstructed-level SPICE kernels. This can be explained by the fact that the smithed-level SPICE kernels related to the test images in Dataset 2 were not generated by an overall adjustment. Thus, compared with reconstructed SPICE kernels, the discrepancies of some adjacent strips derived from smithed-level SPICE kernels were not smaller, but larger. This can explain why the number of matched control measures did not increase significantly after the search window size changed from 31×31 to 51×51 and 71×71 . In other words, a small search window size of 31×31 only works well for partial images whose related smithed-level SPICE kernels were obtained from the same adjustment, while the larger search window size of 71×71 failed to match more conjugate points for all images whose related smithed-level SPICE kernels obtained from different bundle adjustment results.

To illustrate the improvement in image matching success rate with high-resolution LOLA DEM, we provide a comparative matching results between our method and the ISIS's method. The image matching experiment were conducted based on Dataset 1 using a search window size of 51×51 . Using a stereopair composed of M110779506LE and M110799874RE as an example, the image matching results obtained from different methods are shown in Fig. 14.

E. Accuracy Evaluation of Refined Control Measures

In order to further verify the accuracy of the refined control measures obtained using our method, we conducted both

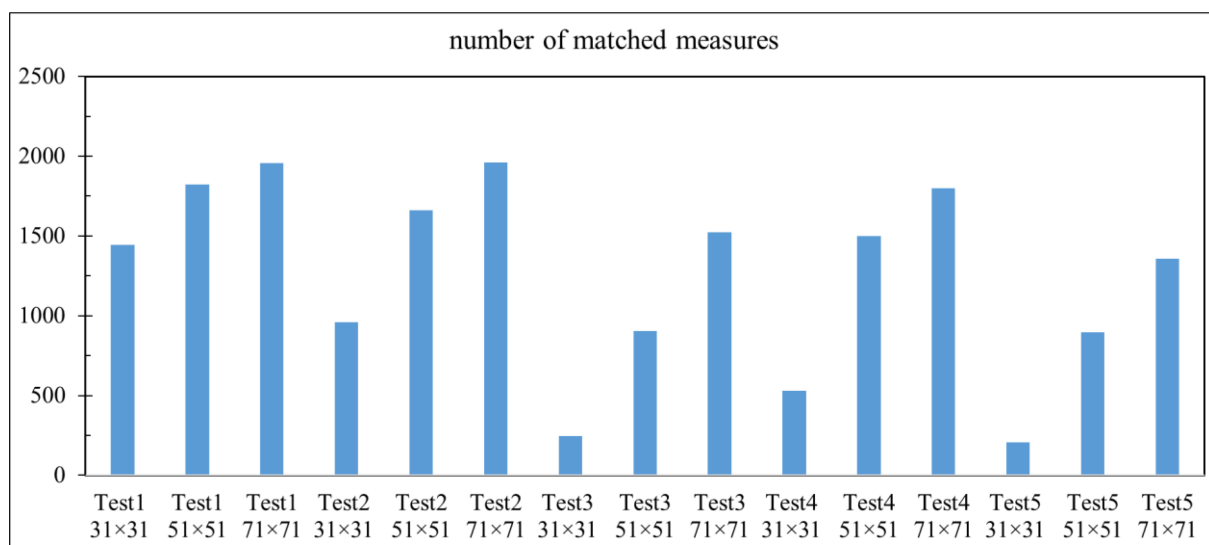


Fig. 12. Number of matched measures based on our method.

TABLE IV
NUMBER OF MATCHED CONTROL MEASURES BY OUR METHOD

Tests	Dataset	Grid spacing of LOLA DEM	Number of candidate measures	Pattern window size	Correlation threshold	Search window size	Number of matched control measures
Test1	Dataset 1	10 m/pixel	2024	19×19	0.8	31×31	1445
						51×51	1824
						71×71	1958
Test2	Dataset 1	30 m/pixel	2024	19×19	0.8	31×31	961
						51×51	1660
						71×71	1959
Test3	Dataset 1	60 m/pixel	2024	19×19	0.8	31×31	244
						51×51	904
						71×71	1524
Test4	Dataset 2	30 m/pixel	5578	19×19	0.8	31×31	528
						51×51	1500
						71×71	1801
Test5	Dataset 2	60 m/pixel	5578	19×19	0.8	31×31	207
						51×51	895
						71×71	1359

TABLE V
NUMBER OF MATCHED CONTROL MEASURES BASED ON “CNETREF” WITH DIFFERENT ACCURACY LEVEL SPICE KERNELS

Dataset	Reconstructed SPICE kernels			Smithed SPICE kernels		
	Number of candidate measures	Search window size	Number of matched control measures	Number of candidate measures	Search window size	Number of matched control measures
Dataset 1	2024	31×31	527	2109	31×31	2044
		51×51	693		51×51	2045
		71×71	1959		71×71	2046
Dataset 2	5578	31×31	1318	5107	31×31	1660
		51×51	2015		51×51	1676
		71×71	2031		71×71	1704

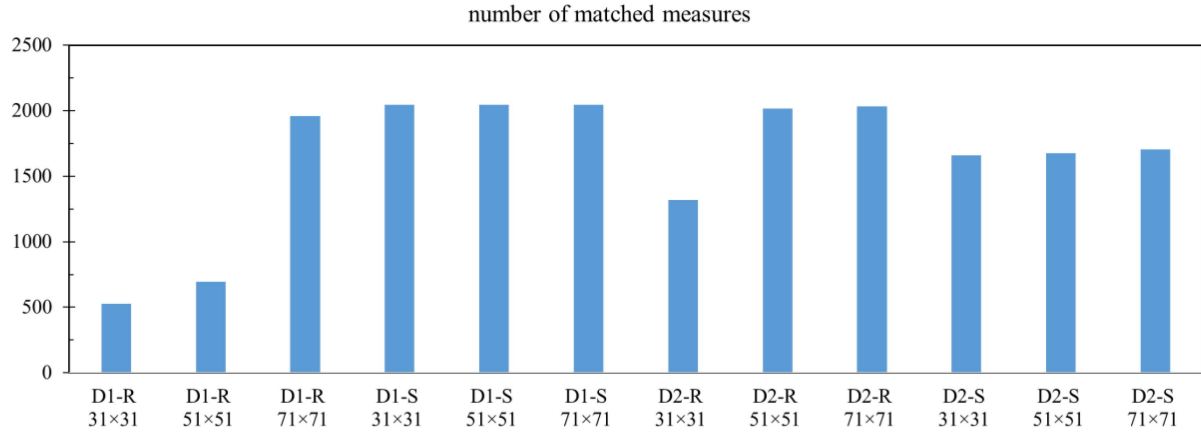


Fig. 13. Number of matched measures based on “cnetref.” D1-R 31×31 indicates Dataset1 with reconstructed level SPICE kernels; D1-S 31×31 indicates Dataset1 with smithed level SPICE kernels. The indicative meanings of other symbols can be inferred in this way.

TABLE VI
QUANTITATIVE ACCURACY EVALUATION RESULTS FOR REFINED CONTROL MEASURES USING A STEREOPAIR COMPOSED OF M110779506LE AND M110820244LE IMAGES

Methods	Pixel coordinates errors (pixels)	
	Sample	Line
ISIS’s cnetref	18.92	-14.82
Our method (10m/pixel DEM)	0.62	-7.33
Our method (30m/pixel DEM)	-5.88	-14.75
Our method (60m/pixel DEM)	16.20	43.12

Note: The control measures’ pixel coordinates errors were computed based on the refined pixel coordinates and the exact pixel coordinates.

qualitative and quantitative accuracy evaluation experiments. We present the control measures generated by different methods using a stereopair composed of M110779506LE and M110820244LE images in Fig. 15 to show the qualitative accuracy evaluation results. To better present the control measures’ accuracy, both the reference and match images are enlarged four-fold. The quantitative accuracy evaluation results were acquired by computing the errors between the refined pixel coordinates based on our method and the exact pixel coordinates determined through image matching (see Table VI). As can be seen, both the qualitative and quantitative evaluation results demonstrate that our method, with the application of 10 and 30 m/pixel LOLA DEMs, can provide more accurate positioning than USGS ISIS’s default processing flow based on reconstructed-level SPICE kernels. This is consistent with the results obtained by ISIS’s “pointreg” matching experiments.

To further evaluate the effectiveness of the proposed control network generation method, a local area of the generated control network for Test 4 is presented in Fig. 16 using a stereopair composed of M153386201RE and M153379413LE. As can be seen, the control points are evenly distributed in the overlapping area. The enlarged subfigures demonstrate that the control points

exhibit high image matching accuracy, and the feature points near the shadows can be correctly matched.

F. Mapping Products Accuracy Evaluation

The sigma naught of the bundle adjustment was 0.48. The reprojection error of most control measures is less than 1 pixel, and the root mean square error of the images in the bundle adjustment is 0.43–0.72 pixels. With the refined exterior orientation parameters obtained through bundle adjustment, we generated mapping products for scientific analysis of the safety assessment of landing sites. We selected the bundle adjustment result obtained from Test 4 with a search window size of 51×51 to generate DOMs with a spatial resolution of 1 m/pixel. The DOMs were generated using ISIS’s “cam2map” program. Fig. 17 presents the relative geometric accuracy of two adjacent image strips. As can be seen, the discrepancies between the two adjacent strips were very small. Based on manually measuring the same feature points on overlapping images, the geometric accuracy of the generated orthophotos is 0.53 pixels. This means that a satisfactory bundle adjustment result was achieved with the high-quality control network generated by our method. This also demonstrates the proposed method’s feasibility.

V. DISCUSSION

As can be seen in Fig. 10 and Table III, for the 10 m/pixel LOLA DEM, the pixel coordinate difference is less than two pixels. For the 30 and 60 m/pixel LOLA DEMs, the pixel coordinate difference is less than 1 pixel. The standard deviation values of the pixel coordinate differences are 0.67 pixels in the sample direction and 0.58 pixels in the line direction for the 10m/pixel LOLA DEM; 0.53 pixels in the sample direction and 0.48 pixels in the line direction for the 30m/pixel LOLA DEM; and 0.22 pixels in the sample direction and 0.29 pixels in the line direction for the 60 m/pixel LOLA DEM. Converting the standard deviation values of the pixel coordinate differences into absolute units in meters shows that the values are between 5 and 20 m. This demonstrates that the proposed method has high prediction accuracy for determining the candidate control

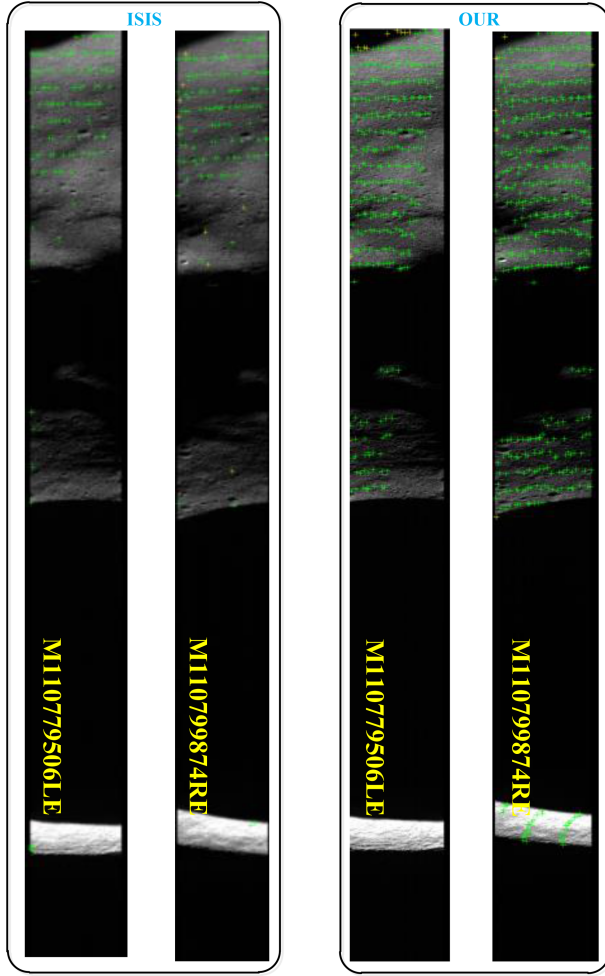


Fig. 14. Image matching results between our method and ISIS's method. The image matching result derived from ISIS's method is based on reconstructed-level SPICE kernels, whereas the image matching result derived from our method is based on shaded DEM maps with 10 m/pixel LOLA DEM.

measures. This lays a good foundation for matching control measures.

We used two sets of LRO NAC images located at the LSP to validate the proposed method. Dataset 1 is closer to the South Pole and spans a latitude range of 87.5° to 90° south latitude. Dataset 2 spans a latitude range of 79.5° to 81.5° south latitude. Dataset 1 contains a large permanent shadow area and numerous shadows, and there are relatively few effective image ranges that can be used for constructing control network. Dataset 2 has some small impact craters and better illumination conditions compared to Dataset 1, but there are obvious differences in image appearance, making it difficult to directly match images.

But we need to emphasize that our method does not necessarily have higher accuracy than ISIS's default method in terms of image matching success rate. In case of our method, the image matching success rate depends on the resolution and accuracy of the LOLA DEM used for rendering shaded DEM maps. In case of ISIS's default method, the image matching success rate mainly depends on the accuracy of SPICE kernels. According to the experimental results, when using 10 and 30 m/pixel LOLA

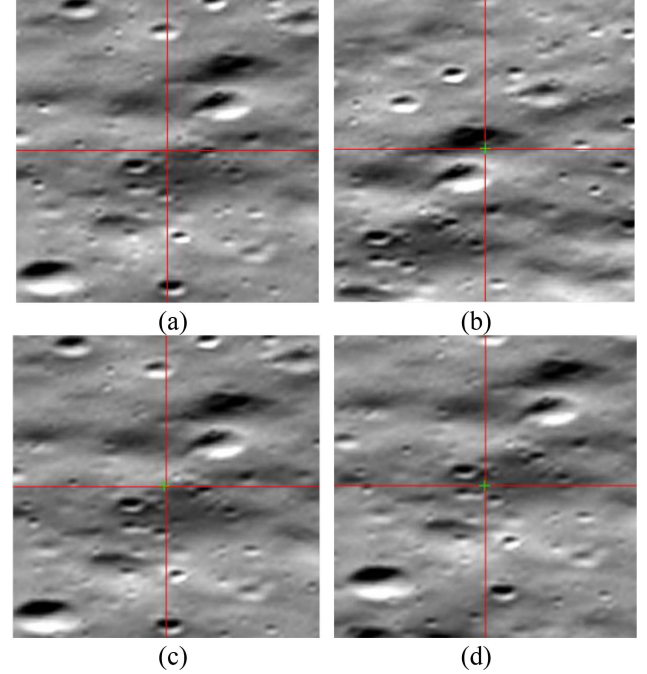


Fig. 15. Comparisons between the default position generated by ISIS's "cnetref" program and the refined position generated by our method using 10 and 30 m/pixel LOLA DEMs, respectively. (a) M110779506LE (reference control measure). (b) M110820244LE, ISIS's method with reconstructed level SPICE kernels. (c) M110820244LE, our method with 10 m/pixel LOLA DEM. (d) M110820244LE, our method with 30 m/pixel LOLA DEM.

DEMs, the predicted accuracy of the control measures based on our method is higher than that provided by reconstructed-level SPICE kernels. Specifically, based on a 10 m/pixel LOLA DEM, the prediction accuracy of control measures can reach several pixels, and based on a 30 m/pixel LOLA DEM, the prediction accuracy of control measures can reach more than ten pixels. This is very advantageous for the photogrammetric processing of lunar south pole images. It is noted that the 10 m/pixel optimized LOLA DEM can cover areas above 83° south latitude, and the 30 m/pixel LOLA DEM can cover areas above 75° south latitude [28], [30]. These areas are currently the most difficult regions for lunar mapping.

Even if the prediction accuracy of the control measures derived from the proposed method based on the LOLA DEM with 60 m/pixel is lower than that obtained directly from the reconstructed-level SPICE kernels, it is still meaningful for the photogrammetric processing of lunar images, especially for those missions whose orbit determination accuracy is lower than that of LRO missions. It is noted that many space agencies plan to obtain higher-resolution remote sensing images covering the entire Moon to derive higher resolution mapping products to support the upcoming exploration missions and scientific research.

As can be seen in Tables IV and V, a larger search window usually obtains more matching points for each dataset. However, the difference in the number of matched control measures can also reflect the relationship between the accuracy of the proposed method and the resolution of the LOLA DEM. Specifically, for

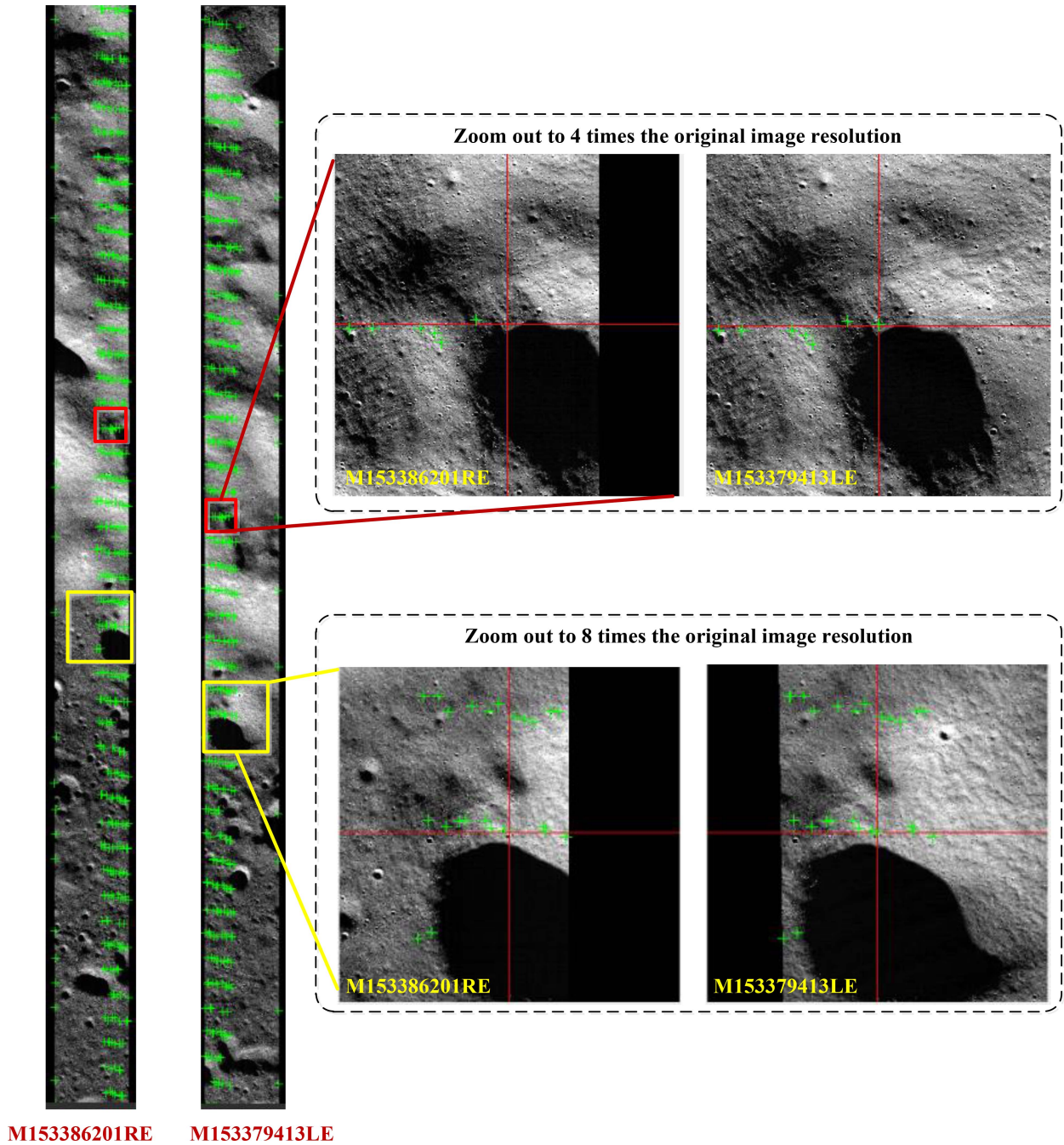


Fig. 16. Illustration of a local area for the generated control network of Test 4. The yellow and red rectangles in the left part of the figure mark the corresponding areas shown in the right part of the figure. The green cross indicates control points.

the 30 m/pixel LOLA DEM in Table IV, it can be seen that a search window size of 51×51 can derive sufficient control measures. Note that the pixel coordinate deviation between the refined control measures and the exact control measures should be less than $(51-19)/2 = 16$ pixels for a search window size of 51×51 . This means that with the application of a 30 m/pixel LOLA DEM, the proposed method can provide a refined control measure with accuracy better than 16 pixels. This accuracy is basically the same as the prediction accuracy obtained from reconstructed-level SPICE kernels for most cases. For tests conducted with 60 m/pixel LOLA DEM, it can be seen that a search

window size of 31×31 obtains very few control measures. This indicates that the prediction accuracy of the proposed method based on the 60 m/pixel LOLA DEM cannot reach the level of a few pixels. A search window size of 51×51 for tests conducted with 60 m/pixel LOLA DEM can obtain more control measures than a search window size of 31×31 , but obviously less than the same search window size for tests conducted with the 10 and 30 m/pixel LOLA DEMs. The search window size of 71×71 obtained the highest number of control measures in all tests. However, it should be noted that, for a search window size of 71×71 , the number of matched control measures for Test 3 is

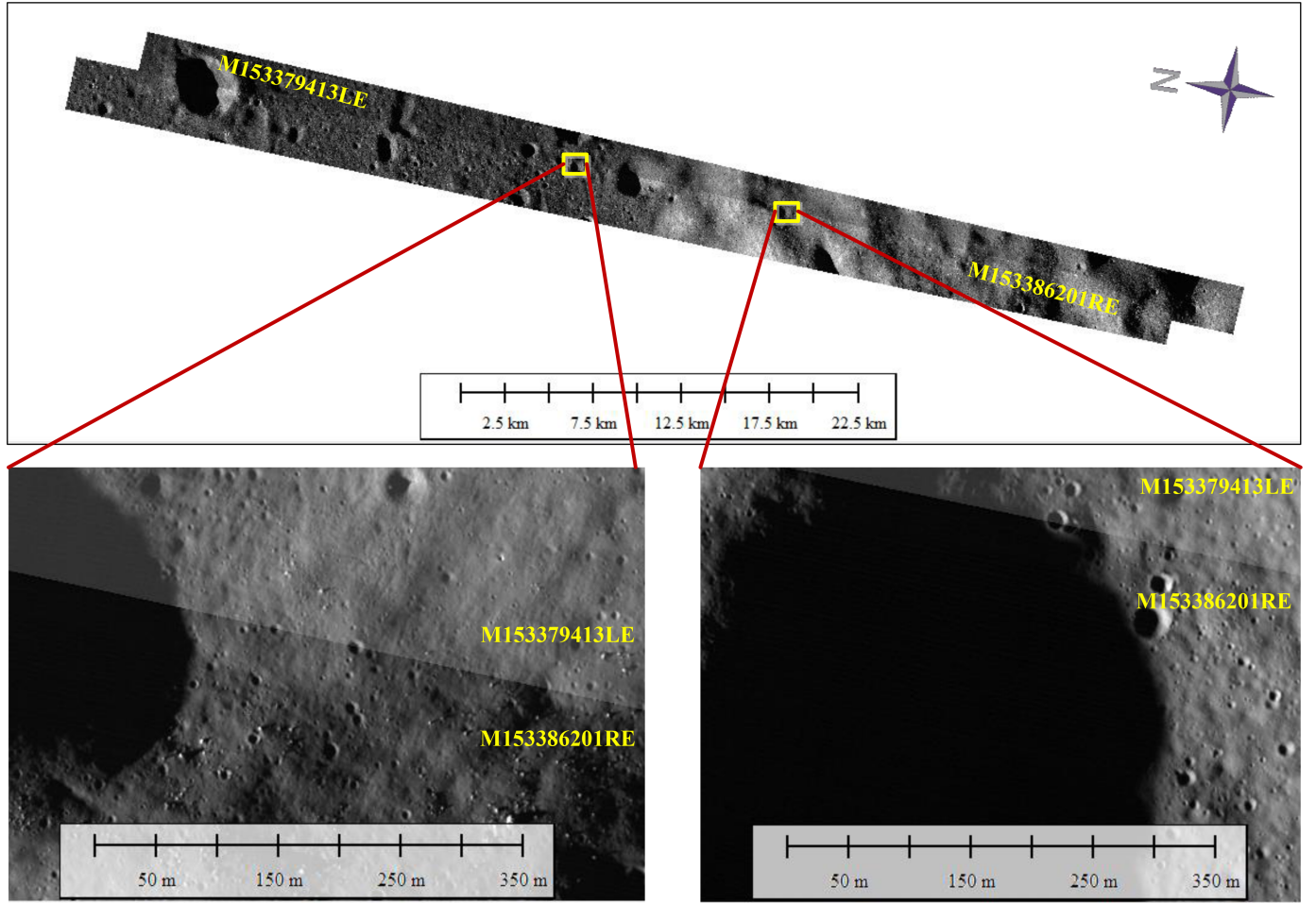


Fig. 17. Geometric accuracy evaluation of the generated DOMs. The bottom part of the figure is shown using a 1:1 scale.

obviously less than those for Tests 1 and 2, and the number of matched control measures for Test 5 is obviously less than that for Test 4. This indicates that the prediction accuracy based on 60 m/pixel may be tens of pixels, which is lower than that of reconstructed-level SPICE kernels.

In short, with the available LOLA DEMs, the proposed method can generate sufficient control measures even if the test images contain a large number of shadows and have poor illumination conditions.

VI. CONCLUSION

The harsh environments such as the low obliquity of the Moon and the heavily cratered terrain in the LSP make image matching and control network construction extremely difficult. A high-quality control network is essential for subsequent photogrammetric processing and mapping product generation and affects the terrain models generated using both SPG and SfS methods. However, existing image matching and control network generation methods perform poorly for lunar south pole images. The image matching of LRO NAC images is significantly influenced by the terrain and illumination conditions in the LSP. The success rate of direct matching on LRO NAC images is very low. In this article, we proposed a new image

matching and control network construction method to address the significant differences in illumination conditions and appearance in lunar south pole images. The proposed method achieved robust matching for all stereopairs of LRO NAC images by using shaded DEM maps as a transition. Since shaded DEM maps are rendered using the same acquisition time of on-orbit LRO NAC images, the illumination conditions of the shaded DEM images will be very consistent with the on-orbit images. Thus, the appearance of their images has become more similar, and the success rate of image matching for the lunar south pole images is greatly improved. Based on the reliable image matching results obtained between the LRO NAC DOMs and the shaded DEM maps, we proposed a new candidate control measure refinement method. It provides a relatively accurate starting position for matching candidate control measures. The approach enables the application of a small search window for image matching while ensuring matching accuracy and efficiency. Consequently, this can significantly enhance the quality of the automatically generated control network.

In addition, the proposed method can be used to add height control points from LOLA DEMs into the control network. As previously published papers have already effectively addressed this issue [27], it will not be discussed further here. There is still room for improvement in the proposed method. For example,

when rendering shaded DEM maps, the shadow distribution and illumination effects may slightly differ from real on-orbit images due to the use of cropped DEM chips, as these cropped DEMs cannot determine whether there are any high mountains causing obstructions beyond their range. Optimization in this regard will be our next research focus, but currently, the matching results obtained using the proposed method can meet the requirements for constructing high-quality control networks in the LSP. Furthermore, the proposed method can provide a means of multisource data fusion processing that can handle the image matching difficulties caused by large differences in illumination and appearance in multi-source images.

REFERENCES

- [1] M. E. Evans and L. D. Graham, "A flexible lunar architecture for exploration (FLARE) supporting NASA's artemis program," *Acta Astronautica*, vol. 177, pp. 351–372, 2020.
- [2] H. M. Yu et al., "Mission analysis and spacecraft design of Chang'E-7," *J. Deep Space Exploration*, vol. 10, no. 6, pp. 567–576, 2023.
- [3] E. J. Speyerer and M. S. Robinson, "Persistently illuminated regions at the lunar poles: Ideal sites for future exploration," *Icarus*, vol. 222, no. 1, pp. 122–136, 2013.
- [4] J. Biswas et al., "Searching for potential ice-rich mining sites on the Moon with the Lunar Volatiles Scout," *Planet. Space Sci.*, vol. 181, 2020, Art. no. 104826.
- [5] P. Hayne et al., "Evidence for exposed water ice in the Moon's south polar regions from lunar reconnaissance orbiter ultraviolet albedo and temperature measurements," *Icarus*, vol. 255, pp. 58–69, 2015.
- [6] P. Gläser et al., "Illumination conditions at the lunar poles: Implications for future exploration," *Planet. Space Sci.*, vol. 162, pp. 170–178, 2018.
- [7] Y. J. Feng, H. T. Li, and X. H. Tong, "Optimized landing site selection at the Lunar South Pole: A convolutional neural network approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, no. 5, pp. 10998–11015, May 2024.
- [8] S. J. Boazman, D. Heather, H. Inoue, and M. Ohtake, "Mapping potential hazards and identifying sampling points within Lunar South Polar region areas of interest," *Planet. Sci. J.*, vol. 5, no. 12, Dec. 2024, Art. no. 279.
- [9] J. Flahaut et al., "Regions of interest (ROI) for future exploration missions to the Lunar South Pole," *Planet. Space Sci.*, vol. 180, 2020, Art. no. 104750.
- [10] M. P. Golombek et al., "Selection of the InSight landing site," *Space Sci. Rev.*, vol. 211, pp. 5–95, 2017.
- [11] K. C. Di et al., "Geospatial technologies for Chang'e-3 and Chang'e-4 lunar rover missions," *Geo-spatial Inf. Sci.*, vol. 23, no. 1, pp. 87–97, 2020.
- [12] R. L. Kirk et al., "Ultrahigh resolution topographic mapping of Mars with MRO HiRISE stereo images: Meter-scale slopes of candidate Phoenix landing sites," *J. Geophys. Res. Planet.*, vol. 113, no. E3, pp. 1–31, 2008.
- [13] W. C. Liu and B. Wu, "Atmosphere-aware photoclinometry for pixel-wise 3D topographic mapping of Mars," *ISPRS J. Photogrammetry Remote Sens.*, vol. 204, pp. 237–256, 2023.
- [14] D. E. Smith et al., "Initial observations from the Lunar Orbiter Laser Altimeter (LOLA)," *Geophys. Res. Lett.*, vol. 37, pp. 1–6, 2010.
- [15] Z. Li et al., "Photogrammetric processing of Tianwen-1 HiRIC imagery for precision topographic mapping on Mars," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4601916.
- [16] B. Wu, W. C. Liu, A. Grumpe, and C. Wöhler, "Construction of pixellevel resolution DEMs from monocular images by shape and albedo from shading constrained with low-resolution DEM," *ISPRS J. Photogrammetry Remote Sens.*, vol. 140, pp. 3–19, 2018.
- [17] P. Liu et al., "The generation of high-resolution mapping products for the lunar south pole using photogrammetry and photoclinometry," *Remote Sens.*, vol. 16, no. 12, 2024, Art. no. 2097.
- [18] Z. Chen, B. Wu, and W. C. Liu, "Mars3dnet: CNN-based high-resolution 3d reconstruction of the Martian surface from single images," *Remote Sens.*, vol. 13, no. 5, 2021, Art. no. 839.
- [19] Y. Tao et al., "Subpixel-scale topography retrieval of Mars using single-image DTM estimation and super-resolution restoration," *Remote Sens.*, vol. 14, no. 2, 2022, Art. no. 257.
- [20] H. Chen, "Neural implicit shape modeling for small planetary bodies from multi-view images using a mask-based classification sampling strategy," *ISPRS J. Photogrammetry Remote Sens.*, vol. 212, pp. 122–145, 2024.
- [21] J. Zhong, J. Yan, M. Li, and J. P. Barriot, "A deep learning-based local feature extraction method for improved image matching and surface reconstruction from Yutu-2 PCAM images on the Moon," *ISPRS J. Photogrammetry Remote Sens.*, vol. 206, pp. 16–29, 2023.
- [22] M. S. Robinson et al., "Lunar Reconnaissance Orbiter Camera (LROC) instrument overview," *Space Sci. Rev.*, vol. 150, pp. 81–124, 2010.
- [23] M. R. Henriksen et al., "Extracting accurate and precise topography from LROC Narrow Angle Camera stereo observations," *Icarus*, vol. 283, pp. 122–137, 2017.
- [24] D. Smith et al., "The Lunar Orbiter Laser altimeter investigation on the Lunar Reconnaissance Orbiter mission," *Space Sci. Rev.*, vol. 150, pp. 209–241, 2010.
- [25] E. Mazarico et al., "Orbit determination of the Lunar Reconnaissance Orbiter," *J. Geodesy*, vol. 86, no. 3, pp. 193–207, 2012.
- [26] M. K. Barker et al., "A new lunar digital elevation model from the Lunar Orbiter Laser Altimeter and SELENE terrain Camera," *Icarus*, vol. 273, pp. 346–355, 2015.
- [27] X. Xin et al., "High-precision co-registration of orbiter imagery and digital elevation model constrained by both geometric and photometric information," *ISPRS J. Photogrammetry Remote Sens.*, vol. 144, pp. 28–37, 2018.
- [28] M. K. Barker et al., "Improved LOLA elevation maps for south pole landing sites: Error estimates and their impact on illumination conditions," *Planet. Space Sci.*, vol. 203, 2021, Art. no. 105119.
- [29] H. Xie et al., "Using laser altimetry to finely map the permanently shadowed regions of the LSP using an iterative self-constrained adjustment strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, no. 9, pp. 9796–9808, Sep. 2022.
- [30] M. K. Barker et al., "A new view of the LSP from the Lunar orbiter laser altimeter (LOLA)," *Planet. Sci. J.*, vol. 4, 2023, Art. no. 183.
- [31] X. Ren et al., "A global adjustment method for photogrammetric processing of Chang'E-2 stereo images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6832–6843, Sep. 2019.
- [32] C. L. Li et al., "Lunar global high-precision terrain reconstruction based on Chang'e-2 stereo images," *Geomatics Inf. Sci. Wuhan Univ.*, vol. 43, no. 4, pp. 485–495, 2018.
- [33] M. Kato et al., "The Kaguya mission overview," *Space Sci. Rev.*, vol. 154, pp. 3–19, 2010.
- [34] M. Ohtake et al., "Performance and scientific objectives of the SELENE (KAGUYA) multiband imager," *Earth Planet. Space*, vol. 60, pp. 257–264, 2018.
- [35] H. Hu and B. Wu, "Block adjustment and coupled epipolar rectification of LROC NAC images for precision lunar topographic mapping," *Planet. Space Sci.*, vol. 160, pp. 26–38, 2018.
- [36] B. Wu and W. C. Liu, "Calibration of boresight offset of LROC NAC imagery for precision lunar topographic mapping," *ISPRS J. Photogrammetry Remote Sens.*, vol. 128, pp. 372–387, 2017.
- [37] X. Tong et al., "Illumination robust landing point visual localization for lunar lander with high-resolution map generation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, no. 1, pp. 1577–1591, Dec. 2025.
- [38] W. C. Liu and B. Wu, "An integrated photogrammetric and photoclinometric approach for illumination-invariant pixel-resolution 3D mapping of the lunar surface," *ISPRS J. Photogrammetry Remote Sens.*, vol. 159, pp. 153–168, 2020.
- [39] Y. Zhang et al., "Analysis of illumination conditions in the lunar south polar region using multi-temporal high-resolution orbital images," *Remote Sens.*, vol. 15, no. 5691, 2023, Art. no. 5691.
- [40] Z. Ye et al., "Comparison and evaluation of feature matching methods for multisource planetary remote sensing imagery," *Photogrammetry Rec.*, vol. 39, no. 7, pp. 845–875, 2024.
- [41] D. DeTone, T. Malisiewicz, and A. Rabinovich, "Superpoint: Self-supervised interest point detection and description," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 224–236.
- [42] P. E. Sarlin et al., "SuperGlue: Learning feature matching with graph neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 4938–4947.
- [43] C. Chen et al., "Large-scale block bundle adjustment of LROC NAC images for LSP mapping based on topographic constraint," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, no. 1, pp. 2731–2746, Dec. 2024.
- [44] L. Xue et al., "Effective feature matching of high-resolution planetary orbiter images based on optimized image partitioning and rapid local correspondence," *Planet. Space Sci.*, vol. 260, 2025, Art. no. 106091.
- [45] B. Wu, H. Zeng, and H. Hu, "Illumination invariant feature point matching for high-resolution planetary remote sensing images," *Planet. Space Sci.*, vol. 152, pp. 45–54, 2018.

- [46] R. Huang et al., "Fast double-channel aggregated feature transform for matching planetary remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, no. 4, pp. 9282–9293, Apr. 2024.
- [47] P. Liu et al., "The generation of high-resolution mapping products for the lunar south pole using photogrammetry and photoclinometry," *Remote Sens.*, vol. 16, no. 12, 2024, Art. no. 2097.
- [48] K. L. Edmundson et al., "Jigsaw: The ISIS3 bundle adjustment for extraterrestrial photogrammetry," *ISPRS Ann. Photogrammetry Remote Sens. Spatial Inf. Sci.*, vol. I-4, pp. 203–208, 2012.
- [49] R. A. Beyer et al., "The Ames Stereo Pipeline: NASA's open source software for deriving and processing terrain data," *Earth Space Sci.*, vol. 5, pp. 537–548, 2018.
- [50] D. E. Shean et al., "An automated, open-source pipeline for mass production of Digital Elevation Models (DEMs) from very-high-resolution commercial stereo satellite imagery," *ISPRS J. Photogrammetry Remote Sens.*, vol. 116, pp. 101–117, 2016.
- [51] B. K. P. Horn, "Hill shading and the reflectance map," *Geo-Processing*, vol. 2, no. 1, pp. 65–146, 1982.
- [52] C. Acton et al., "SPICE tools supporting planetary remote sensing," *ISPRS - Int. Arch. Photogrammetry Remote Sens. Spatial Inf. Sci.*, vol. XLI-B4, pp. 357–359, 2016.
- [53] X. Geng et al., "A novel pixel-level image matching method for Mars Express HRSC linear pushbroom imagery using approximate orthophotos," *Remote Sens.*, vol. 9, no. 12, 2017, Art. no. 1262.
- [54] X. Geng et al., "Generation of large-scale orthophoto mosaics using MEX HRSC images for the candidate landing regions of China's first Mars mission," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, no. 1, Jan. 2022, Art. no. 5613520.
- [55] X. Geng et al., "A robust ground-to-image transformation algorithm and its applications in the geometric processing of linear pushbroom images," *Earth Space Sci.*, vol. 6, no. 10, pp. 1805–1830, 2019.



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